

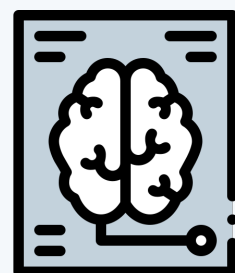
DATA



Datasets

HCP
CamCAN
Abide II

=



+

Targets

Gender
Age
DX Group

PREPROCESSING



Denoising → Distorsion Correction → T1w preprocessing → DWI-T1w registration

DATA PREPARATION



Tissue
type

Gray matter



White matter

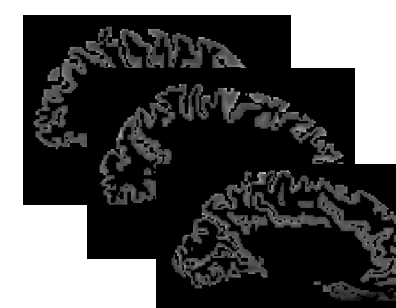


Microstructure

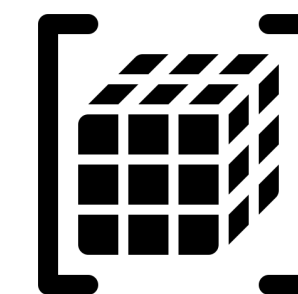
- b0
- mean-diffusivity
- mean-kurtosis
- spherical harmonics

Data representation

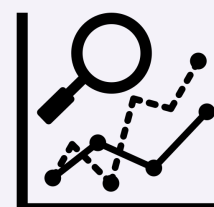
3D tensor



Array



PREDICTION



Evaluation Setup

Stratified KFold
Controlled Seed
Cross validation

Models

Array based

PCA + lin. reg.
Lasso
PCA + SVM
Rand. Forest

Deep Learning

MedicalNet
DinoV2
Curia

Utils



- Parallelization for data prep. and training (joblib, SLURM)
- CLI commands for preparation/training/analysis
- Hydra-based configuration
- Caching for features/embeddings/results
- BIDS-compatible, including multi-center